T7.2.3-Worksheet 2-Pseudocode-Count-Controlled Loop

Answer the following questions.

1. Write pseudocode using a `FOR` loop that outputs the message "`Welcome to Computer Science`" exactly 10 times.

2. Write pseudocode using a `FOR` loop that asks the user to enter 5 numbers one by one, calculates the total sum of these 5 numbers, and outputs the final total.

3. Write pseudocode to input the test scores for a fixed group of exactly 8 students. For each score entered, determine if it is 50 or above, and if so, record it as a pass. Once all 8 scores have been processed, output the final number of students who passed. Declare all variables used.

4. Write pseudocode that prompts the user to enter a single whole number. Using a `FOR` loop, output the multiplication table for that number from 1 to 12 (e.g., displaying `Number * 1, Number * 2`, up to `Number * 12`).

5. A weather station records the daily temperature in Celsius at 12:00 PM every day for a week (7 days).

   Write pseudocode using a `FOR` loop that:

   a. Inputs the daily temperature for each of the 7 days.
   b. Calculates the total sum and the average temperature for the week.
   c. Determines and stores the highest recorded temperature of the week.
   d. Outputs the calculated average temperature and the highest temperature.

   Declare all variables used with appropriate data types.

6. A bookstore processes orders for a fixed batch of exactly 15 customers. Each customer enters the total cost of their book purchase.

Write an algorithm in pseudocode to:

a. Input the order amount for each of the 15 customers.
b. Calculate the total revenue generated from all 15 orders.
c. Count how many customers qualify for free gift wrapping (orders of **$50 or more**).
d. Count how many customers placed small orders under **$20**.
e. Output the total revenue, the number of qualifying gift wrap orders, and the number of small orders after all 15 customer purchases are entered.

Declare all variables used with appropriate data types.

7. Challenge Yourself Question:

A construction plant manufactures a pre-ordered batch of exactly 20 solid concrete pillars shaped as right trapezoidal prisms. Write an algorithm in pseudocode that inputs the two parallel bases (A and B), the perpendicular height (H), and the length (L) in meters for each of the 20 pillars, validating that all entered measurements are greater than zero before computing the volume using $Volume = \frac{A+B}{2} \times H \times L$. The program must calculate the total concrete volume required for the batch, track the smallest pillar volume produced, count how many individual pillars exceed a standard heavy-duty threshold of $15.0m^3$, and output these three final values once all 20 pillars have been processed. Declare all variables used with standard IGCSE data types.

End of Question Paper